Master's Project Report

CISC 898 002 Master's Project S26

KPI-RAG: Explainable Root-Cause Analysis for 5G Networks via Standards-Grounded Retrieval-Augmented Generation

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1 Abstract

This report presents the design, implementation, evaluation, and deployment of KPI-RAG, a Retrieval-Augmented Generation (RAG) system for explainable 5G network fault diagnosis. The system receives a structured fault prediction from an upstream machine learning classifier and produces a standards-grounded explanation comprising a root cause analysis, SHAP-attributed KPI evidence, a specific 3GPP standard clause reference, and a recommended remediation action.

Exploratory data analysis and the correction of two pipeline defects (silently zero-filled categorical features, and near-duplicate leakage in a row-level split with 31.8% of test windows having a $\geq$0.999-cosine training match, reduced to 4.05%) led to a validated session-level holdout split (300-second gap threshold; 33 sessions; 25,688 training / 6,312 test windows) and a corrected 584-dimensional feature vector. A systematic comparison of four tree-ensemble algorithms across four feature configurations, confirmed by a five-seed robustness check, selected CatBoost with a 64-dimensional statistics-only configuration.

On the final split, the binary detector achieves 0.897 recall at the natural test rate (paper-comparable F1 = 0.698 versus 0.227 for the Isolation Forest baseline), and the 11-class fault classifier achieves 0.677 accuracy with a cross-validated F1-macro of 0.333 ± 0.076 (0.354 ± 0.072 under evidence-based class grouping). A structural failure mechanism is identified and documented: an implicit catch-all class in which 82.2% of detector false positives are labelled Co-Channel Interference (Mild).

A three-condition ablation (Track C) quantified the contribution of 3GPP alignment-table grounding to citation faithfulness, yielding citation validity rates of 7% (LLM baseline, no context), 10% (LLM with ticket retrieval), and 90% (full system with alignment-table grounding), a +80 percentage-point improvement isolating the contribution of standards grounding. An explanation-layer technical audit corrected three further defects, and a test suite of 118 tests (81 unit, 37 integration) including negative-case validation was maintained at 100% pass rate. The system is deployed on Streamlit Community Cloud with a modular six-panel dashboard providing live, on-demand pipeline execution within a $\sim$430 MB memory footprint.

2 Introduction / Background

Fifth-generation (5G) mobile networks operate across a complex hierarchy of radio units (O-RU), distributed units (O-DU), centralised units (O-CU), and near-real-time RAN Intelligent Controllers (Near-RT RIC), all governed by the O-RAN Alliance architecture and the 3GPP standards body. When anomalous key performance indicator (KPI) behaviour is detected, indicating faults such as co-channel interference, Doppler shift degradation,

buffer overflow, or RF jamming, network operations centre (NOC) engineers require more than a fault classification label. They require a causally grounded explanation: why the fault occurred, which 3GPP Technical Specification or Technical Report governs the observed behaviour, and what corrective action is appropriate.

Existing fault detection systems in the telecommunications literature typically terminate at classification. The diagnostic gap between a predicted fault label and an actionable, standards-referenced explanation remains largely unaddressed. Engineers currently correlate raw KPI measurements against technical specifications manually: a process that is slow, inconsistent, and unscalable with fault volume (1). Without automation, every incident depends on individual expertise, prolonging mean-time-to-resolution (MTTR) and increasing operational cost. The reasoning behind diagnostic decisions is rarely captured in a structured, reusable form, making diagnosis difficult to reproduce consistently across similar incidents.

KPI-RAG addresses this gap by connecting KPI-signal-level detection directly to a validated, citation-backed explanation layer within one evaluable pipeline, rather than treating anomaly detection and standards-compliant explanation as separate problems. To the best of our knowledge, no published system addresses all three stages (detection, classification, and standards-grounded explanation) in a single evaluable pipeline on public benchmark data.

Specific technical contributions:

- A ChromaDB-backed vector knowledge base of 935 historically indexed fault tickets from the training partition, retrieved by cosine similarity at inference time with a configurable confidence threshold.
- A multi-layer RCA evidence mechanism incorporating SHAP feature attributions (Layer B), KPI observational statistics (Layer A), and causal mechanism descriptions with standards references (Layer C) from a pre-computed evidence store, injected into the grounded-explanation prompt and displayed alongside the explanation.
- A curated 3GPP alignment table mapping ten fault types to specific standard clauses, clause text excerpts, and O-RAN architectural components, used to guide and validate LLM citation behaviour.
- A three-condition ablation evaluation framework quantifying the independent contribution of retrieval and alignment components to citation validity.
- A modular Streamlit dashboard providing live pipeline execution with six specialised display panels, deployed on Streamlit Community Cloud.
- A proactive technical audit identifying and remediating three engineering defects that affected evaluation validity and pipeline coherence.

3 Problem Statement

Given a window of 5G KPI measurements, this project addresses three connected sub-problems: (1) detecting whether observations represent anomalous behavior; (2) classifying the fault type among eleven categories (ten synthetically induced, one real jamming event); (3) generating a root-cause explanation grounded in official 3GPP and O-RAN specifications. To the best of our knowledge, no published system addresses all three in a single evaluable pipeline on public data.

Scope boundaries: Single-cell, single-modality (KPI time-series) fault diagnosis on TelecomTS. Multi-cell correlation, cross-domain RCA spanning RAN, core, and transport, and live production deployment are out of scope and noted as future work.

Scope of Validity: KPI-RAG is evaluated on a single indoor 5G testbed with 10 synthetically injected and 1 controlled real anomaly type. Generalization to multi-vendor, multi-cell, multi-operator production deployments is explicitly out of scope. Production deployment would require re-evaluation on operator data with natively occurring fault labels.

Research Questions:

- **RQ1:** Does scale-preserving feature engineering improve anomaly detection F1 and fault classification F1-macro vs. normalized-feature and statistics-only baselines?
- **RQ2:** Does alignment-table grounding produce measurably higher citation faithfulness vs. ticket-only and ungrounded LLM generation?
- **RQ3:** Which feature configuration offers the best performance-to-cost tradeoff?

4 Research Objectives

Primary Objective: Design, implement, and evaluate an end-to-end pipeline that detects and explains 5G network faults with verifiable standards grounding.

- **O1: Anomaly Detection.** Train a scale-aware two-stage pipeline on TelecomTS across 11 fault types from 18 KPI channels, quantifying the contribution of absolute-scale feature preservation. Model 1: binary anomaly detection F1 on TelecomTS. Model 2: per-class F1-macro across all 11 classes.
- **O2: Fault-to-Standard Alignment.** Construct and validate a 10-row Alignment Table linking synthetic fault categories to specific 3GPP clauses and O-RAN components (jamming excluded, no corresponding clause), cross-referenced against GSMA KGs and confirmed against primary specification text.
- **O3: Standards Retrieval.** RAG pipeline using SHAP attributions as the structured

retrieval query. Criterion: acceptable precision@k on held-out alignment-table queries.

- **O4: Grounded Explanation.** Plain-language reports where every claim is grounded in a retrieved clause or ticket. Criterion: $\geq$70% pass the regex-validated 3GPP clause check.
- **O5: Evaluation.** Joint detection and explanation faithfulness protocol. No existing benchmark evaluates both jointly, making this a contribution in itself.

Summary of Contributions: (1) Artifact: first TelecomTS-to-3GPP/O-RAN alignment mapping with independently verified evidence spans for 10 fault types. (2) Empirical finding: Track C ablation measures whether standards grounding produces higher citation faithfulness than ticket-retrieval-only and ungrounded generation. (3) Pipeline demonstrator: first per-class F1 table for the 11-class TelecomTS task, paired with an evaluable end-to-end implementation.

5 Related Work

This section reviews prior work related to KPI-based anomaly detection, telecommunications knowledge retrieval, and large language model (LLM)-based root cause analysis (RCA). The literature is organized into four areas: anomaly detection in 5G networks, telecom knowledge retrieval and retrieval-augmented generation (RAG), LLM-based root cause analysis, and evaluation of LLM-generated explanations. The section concludes by identifying the research gaps addressed by KPI-RAG.

5.1 Anomaly Detection in 5G Networks

Recent research has investigated automated anomaly detection in telecommunications networks using KPI time-series data and machine learning methods. Feofanov et al. (2) introduce Mantis, a lightweight foundation model for time-series classification, and report an F1 score of 0.800 on the binary anomaly detection task using the TelecomTS benchmark. However, the approach focuses on classification and does not provide KPI-level attribution or standards-grounded explanations. The TelecomTS dataset is introduced by Feng et al.. (3) as a multi-modal observability benchmark combining time-series and language information for telecommunications network analysis.

Beyond telecom-specific benchmarks, Lin et al.. (4) propose NuTime, a numerically multi-scaled embedding approach for large-scale time-series pretraining. The present work adopts the general principle of capturing local patch-level statistics while using a different architecture and task formulation. Malacarne et al.. (5) further investigate scalable context-aware graph attention for unsupervised anomaly detection in large-scale mobile networks, demonstrating the applicability of machine-learning-based KPI monitoring in a production

telecom environment.

These studies demonstrate the feasibility of automated anomaly detection in both benchmark and real-world telecommunications settings. However, their primary objective is anomaly detection or classification rather than generating explanations that connect observed KPI behavior to specific network components, technical standards, or operational root causes. This limitation motivates the integration of anomaly detection with standards-grounded RCA considered in KPI-RAG.

5.2 Telecom Knowledge Retrieval and RAG

Large language models have increasingly been applied to telecommunications-specific knowledge tasks, where retrieval can provide access to domain-specific information that may not be reliably encoded in the model parameters. Maatouk et al.. (6) introduce TeleQnA, a benchmark designed to assess the telecommunications knowledge of LLMs. The benchmark demonstrates the importance of domain-specific knowledge and retrieval for answering specialized telecommunications questions.

The TelecomTS benchmark also provides a complementary multimodal setting for combining network time-series information with textual information (3). In addition, publicly available knowledge resources such as the TelecomTS dataset repository (7), the 3GPP Release 19 knowledge graph (8), and the O-RAN specification knowledge graph (9) provide structured sources that can support standards-grounded retrieval.

These resources are relevant to KPI-RAG because they provide the knowledge and standards context required for grounding generated explanations. However, telecom knowledge benchmarks such as TeleQnA primarily evaluate question answering rather than root cause explanation from observed KPI anomalies. Similarly, knowledge graphs provide structured information but do not by themselves establish a complete pipeline from an anomalous KPI window to a specific root cause and supporting standards clause.

5.3 LLM-Based Root Cause Analysis

LLMs have also been explored for automated root cause analysis in cloud and IT operations. Zhang et al.. (1) provide a broad survey of AIOps in the era of LLMs and identify domain grounding, reliable reasoning, and evaluation of generated outputs as important challenges for LLM-based operational diagnosis.

Chen et al.. (10) propose an LLM-based approach for automatic root cause analysis of cloud incidents and demonstrate the effectiveness of combining LLM reasoning with operational information. Although this work shows the potential of LLMs for RCA, its focus is on cloud incidents rather than telecommunications KPI time-series, and it does not provide telecommunications standards-grounded explanations.

Roy et al.. (11) investigate LLM-based agents for root cause analysis and emphasize the importance of evaluating the quality and groundedness of generated RCA outputs. Their work is relevant to KPI-RAG because it supports the use of structured evaluation criteria for assessing LLM-generated diagnostic explanations.

More recently, Tian et al.. (12) propose GALA, a graph-augmented LLM agentic workflow for root cause analysis. Their work demonstrates how structured relationships can be incorporated into agent-based RCA to improve the reasoning process. However, the proposed approach is not specifically designed around telecom KPI anomaly windows or clause-level 3GPP and O-RAN grounding.

In the telecommunications domain, Wu et al.. (13) introduce TN-AutoRCA, a benchmark and agentic framework for alarm-based root cause analysis. While this work directly addresses telecommunications RCA, its primary input modality is network alarms and associated information rather than KPI time-series windows. This creates a different problem formulation from the KPI-based anomaly-to-explanation pipeline considered in this work.

Sana et al.. (14) investigate reasoning language models for root cause analysis in 5G wireless networks and demonstrate the potential of domain-specific reasoning models for telecom RCA. Their setting, however, focuses on a different data modality from the KPI-window formulation adopted by KPI-RAG.

Overall, existing LLM-based RCA studies demonstrate the potential of language models, retrieval, agents, and structured information for operational diagnosis. Nevertheless, the reviewed approaches generally focus on cloud metrics, alarms, graph information, or other network modalities rather than explicitly connecting KPI-level anomalies to standards-grounded root cause explanations.

5.4 Evaluation of LLM-Generated Explanations

Evaluating LLM-generated explanations requires more than conventional classification metrics because an explanation may be factually plausible while lacking relevance, consistency, or adequate grounding. Liu et al.. (15) introduce G-Eval, an evaluation framework that uses LLMs to assess generated natural-language outputs according to predefined criteria and demonstrates improved alignment with human judgments.

The G-Eval framework provides a foundation for evaluating the quality of generated RCA explanations in KPI-RAG. However, generic text-generation evaluation does not directly capture the requirements of telecommunications standards-grounded diagnosis. Therefore, the present work adapts the general evaluation principle to the specific requirements of telecom RCA, including the assessment of explanation quality and standards grounding. The detailed evaluation criteria and scoring procedure are presented in Section 8.

5.5 Research Gaps

The reviewed literature reveals several gaps that motivate KPI-RAG. First, existing telecom anomaly detection studies primarily focus on detecting or classifying anomalous behavior and do not explicitly connect detected KPI patterns to standards-grounded root cause explanations (2; 3; 5). Second, telecom knowledge benchmarks such as TeleQnA evaluate telecommunications knowledge and question answering but do not directly generate root cause explanations from observed network KPI anomalies (6). Third, LLM-based RCA systems demonstrate the potential of retrieval, agents, and language-model reasoning, but their input modalities and application domains differ from the KPI time-series setting considered in this work (10; 11; 12; 13; 14).

The availability of structured telecom knowledge resources, including 3GPP- and O-RAN-oriented knowledge graphs (8; 9), provides an opportunity to address the grounding problem. However, these resources do not themselves define a complete pipeline that begins with KPI anomaly detection and produces a root cause explanation supported by specific standards information.

To the best of our knowledge, the reviewed literature has not yet demonstrated a single evaluable pipeline that connects KPI-level anomaly detection with clause-level, standards-grounded root cause explanation on a public telecom benchmark. In particular, an explicit mapping between TelecomTS fault categories, observed KPI behavior, relevant 3GPP information, and associated O-RAN components is not established within a single evaluation framework in the reviewed works.

KPI-RAG addresses these gaps by integrating KPI anomaly detection, telecommunications knowledge retrieval, standards grounding, and LLM-based root cause explanation into a unified pipeline. The generated explanations are subsequently evaluated using an adapted G-Eval-based framework, with project-defined evaluation dimensions appropriate to the telecommunications standards-citation setting. The detailed methodology is presented in Section 8.

6 Proposed Approach / Methodology

6.1 High-Level Approach and System Architecture

KPI-RAG is a three-layer pipeline that takes a raw 5G KPI time series as input and produces a standards-grounded fault diagnosis as output. The three layers run sequentially during inference. When the binary detector outputs Normal, the pipeline terminates at Layer 2 and displays a "No anomaly detected" message; Layer 3 is not invoked.

Layer 1: Preprocessing and Feature Engineering. Loads TelecomTS at native scale;

constructs the 584-dimensional feature vector per sample; exports a session-level holdout train/test split as shared arrays for both models.

Layer 2: Anomaly Detection, Fault Classification, and SHAP. Two cascaded CatBoost models (a binary detector followed by an 11-class fault classifier) with SHAP TreeExplainer producing per-channel attribution scores. Output: structured JSON consumed by Layer 3.

Layer 3: RAG and Standards-Grounded Explanation. Retrieves top-5 similar historical tickets from TelecomTS (ChromaDB), injects the pre-validated 3GPP clause and O-RAN component from the Alignment Table, and generates a structured explanation via an open-source LLM. Results are presented on a Streamlit dashboard deployed on Streamlit Community Cloud.

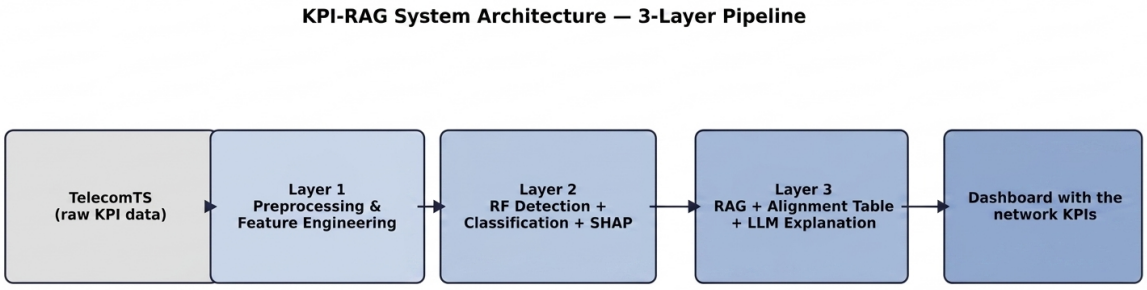


Figure 1. KPI-RAG System Architecture: 3-Layer Pipeline. Raw KPI time-series data flows through Layer 1 (preprocessing and feature engineering), Layer 2 (anomaly detection, fault classification, and SHAP), and Layer 3 (RAG and standards-grounded explanation), culminating in a dashboard displaying the fault diagnosis.

6.2 Dataset: TelecomTS

TelecomTS (3) provides 32,000 labeled samples recorded from a real 5G lab network at 100 ms resolution across 18 KPI channels spanning PHY, MAC, and network layers. The dataset contains 30,765 normal samples and 1,235 anomalous samples across 11 fault categories: one real over-the-air jamming event captured by a USRP X310 jammer, and 10 synthetic fault types injected into normal recordings following KPI-effect mappings validated against the telecom engineering literature.

6.3 Exploratory Data Analysis

Before designing the feature vector or selecting a train/test split strategy, the full TelecomTS dataset was explored to understand its statistical properties, temporal structure, and potential modelling risks. This analysis was directed at four specific questions: how

Table 1. TelecomTS dataset properties.

Property	Value
Total samples	32,000
Normal samples	30,765 (96.1%)
Anomalous samples	1,235 (3.9%)
KPI channels	18 (16 numerical, 2 categorical)
Timesteps per window	128 at 100 ms resolution = 12.8 seconds
Fault categories	11 (1 real jamming, 10 synthetic)
Troubleshooting tickets	1,235 GPT-4.1-generated diagnostic texts, one per anomalous sample
License	MIT

discriminative are the raw KPI channels, how severe is the class imbalance, how are recording sessions structured in time, and do the metadata labels introduce spurious shortcut features. Each finding directly informed a design decision in the preprocessing and modelling pipeline.

Class imbalance and fault distribution. The dataset is severely imbalanced at 96.1% Normal and 3.9% Anomalous, reflecting the natural rarity of faults in production 5G networks. Within the anomalous class, fault-type counts range from 279 windows (Jamming, the only real over-the-air fault) to 48 windows (High Network Congestion Sudden Spike), a 5.8× spread across eleven categories. This imbalance has two direct modelling consequences: Model 1 uses an 80/20 normal/anomalous training rebalance with F2-score as the tuning objective, prioritising recall over precision because a missed fault is more costly than a false alarm; and the two smallest classes ($n \approx 37$ and $n \approx 43$ after the session split) are evaluated with stratified 5-fold cross-validation rather than a single train/test split.

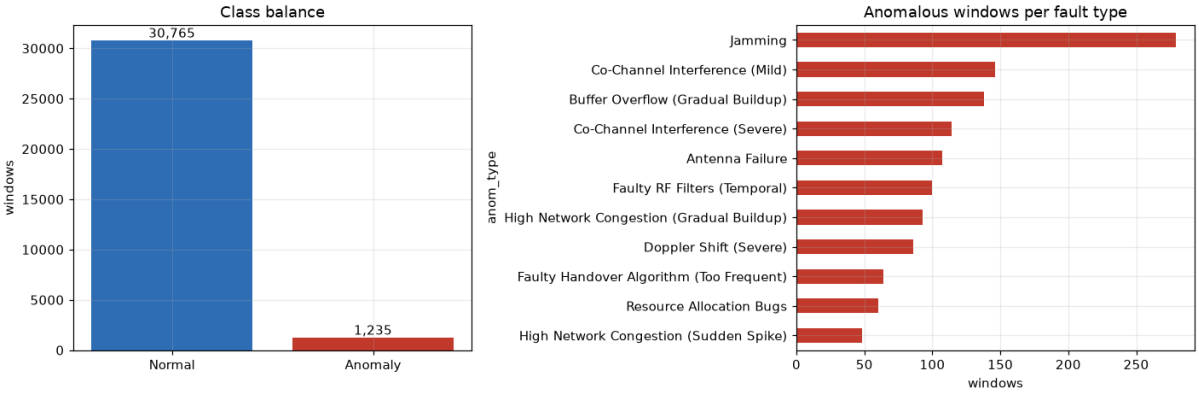


Figure 2. Class balance and fault-type distribution in TelecomTS. Left: overall class imbalance (30,765 Normal vs. 1,235 Anomalous). Right: window count per fault type. Jamming is the most frequent (279); High Network Congestion (Sudden Spike) is the rarest (48).

Channel discriminability. Given the severe class imbalance confirmed above, under-

standing which KPI channels actually separate the two classes is critical for feature design. Comparing per-channel means between Normal and Anomalous windows reveals that `Estimated_UL_Buffer` shows the largest separation (+1,001% mean difference), followed by `PRB_Utilization_UL` (+272%) and `UL_NumberOfPackets` (+265%). RSRP, the standard 5G signal quality indicator, shows only a 4.8% mean difference. This confirms that global per-channel statistics carry genuine discriminative signal and validates their use as the primary feature representation, while cautioning against over-reliance on RSRP as the primary fault indicator. The heatmap below further shows which KPI channels are most frequently associated with Jamming, the only fault type with populated `affected_kpis` metadata, providing a partial ground-truth reference for the SHAP attribution results in Section 7.7.

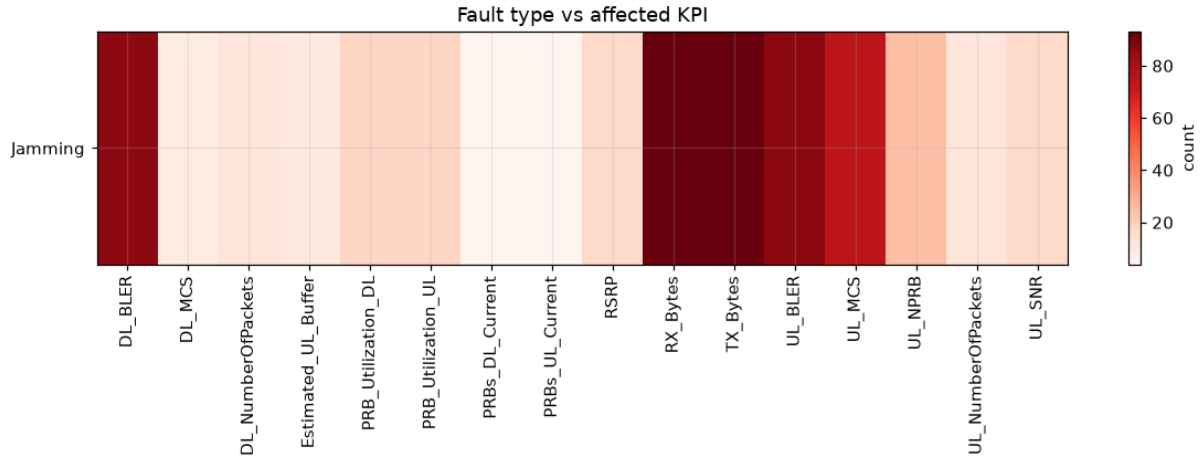


Figure 3. Fault type versus affected KPI heatmap. Only Jamming windows have populated `affected_kpis` metadata; the impact on other fault types is inferred from SHAP analysis (Section 7.7).

Session structure. TelecomTS windows are created with a 75% overlap between adjacent windows, a stride of 32 timesteps over a window of 128. This overlap means that a naive row-level train/test split would allow near-duplicate windows to appear in both partitions, inflating evaluation metrics. To design a clean split, the temporal structure of the dataset was analysed by applying gap threshold sensitivity analysis across nine thresholds from 1 second to 24 hours. Session count stabilises at 300 seconds, yielding 33 sessions versus 32 at 3,600 seconds, with an average session size of approximately 970 windows. This finding directly determined the 300-second session boundary threshold used in the final holdout split, where entire sessions are assigned to either train or test with zero overlap.

Context label shortcuts. The TelecomTS `labels` column contains operational context fields (zone, application, mobility, and congestion) that are not KPI measurements. Inspection reveals that windows labelled `Mobility=Yes` (927 windows) and `Congestion=Yes` (2,781 windows) contain zero anomalous samples. Had these fields been included as classifier inputs, the model could achieve deceptively high accuracy by learning that

`Mobility=Yes` implies Normal, a spurious shortcut rather than a genuine fault pattern. This finding, combined with the label-leakage risk from fields derived from ground-truth anomaly information, validates the decision to exclude the `labels` column entirely from classifier inputs and recompute all statistics from raw KPI values only.

6.4 Layer 1: Preprocessing and Feature Engineering

The core design decision is that no normalisation is applied at any stage. Tree-based classifiers are invariant to monotonic transformations, but absolute-magnitude information must be preserved at the feature-extraction stage: normalising the raw time-series before feature extraction discards the diagnostic signal contained in absolute KPI levels. For example, RSRP = −90 dBm is diagnostically different from RSRP = −70 dBm; this difference must survive into the feature vector. Two defects were identified and corrected during implementation, both of which affected all prior pipeline versions (detailed in Section 7.3).

Each 128-timestep window produces a 584-dimensional feature vector with four components:

Table 2. Feature vector architecture (584 dimensions total).

Component	Dims	How Computed	Why
Precomputed statistics	64 (4×16)	Mean, std, min, max per channel; recomputed from raw KPI values. Metadata fields excluded to prevent label leakage.	Simple and meaningful per-channel summaries computed from the current window only.
Patchwise scale statistics	256 ($8 \times 16 \times 2$)	The 128 timesteps are divided into 8 equal patches. For each patch and each channel, the mean and standard deviation are computed.	Preserves absolute magnitude within each temporal patch; key for scale-aware detection following the NuTime principle (4).
First-order differences	256 ($8 \times 16 \times 2$)	First-order differences are computed over the raw time series. For each of the 8 patches and 16 channels, the mean and standard deviation of the difference signal are extracted.	Captures rate of change and temporal fault progression rather than only absolute signal levels.

Continued on next page

Component	Dims	How Computed	Why
Categorical encoding	8 (4×2)	One-hot encoding of `UL_Protocol` and `DL_Protocol`. Each direction has four categories: TCP, UDP, None, and EMPTY.	Protocol state is diagnostically relevant under fault conditions.
Total	**584**		

Split strategy. The final pipeline uses a session-level holdout split that assigns entire recording sessions to either train or test, eliminating the adjacent-window leakage present in a row-level stratified split. Session boundaries were detected by applying gap thresholds ranging from 1 second to 24 hours to the timestamp sequence. Session count stabilises at 300 seconds, yielding 33 sessions versus 32 at 3,600 seconds, while producing sessions of sufficient average size (approximately 970 windows each) for reliable per-session evaluation. Smaller thresholds fragment the data into sessions too small for this purpose: at 60 seconds, 62 sessions are detected, many containing fewer than 100 windows. **The 300-second threshold was therefore selected as the session boundary.**

Test sessions {1, 3, 4, 13, 17} were selected to guarantee full fault-type coverage in both partitions. The resulting split contains 25,688 training windows and 6,312 test windows, with a test anomaly rate of approximately 4.75%. **A recording-level split was evaluated and rejected: all 1,235 anomalous windows were concentrated in a subset of sessions, producing a test set with zero anomalous samples.** The row-level stratified split was also evaluated; near-duplicate analysis showed a 31.8% rate of test windows with a $\geq$0.999-cosine-similar training match, falling to 4.05% under the session-level holdout. The same partition indices apply to the RAG corpus: only tickets from training-partition windows are indexed in ChromaDB; the test partition is held out and never indexed.

6.5 Layer 2: Anomaly Detection, Fault Classification, and SHAP

Two cascaded CatBoost classifiers are trained on the 64-dimensional Config A feature vectors produced by Layer 1.

A single 12-class joint classifier (Normal + 11 fault types in one model) was evaluated as an alternative to the cascade. It was rejected for three reasons: the 26:1 class imbalance between Normal and the rarest fault class makes joint training unstable; the two tasks

require different optimisation objectives (recall-first for detection, macro-F1 for classification) that cannot be tuned independently in a single model; and the binary detection threshold cannot be adjusted for downstream consumers without retraining the entire classifier. The cascade architecture resolves all three.

Algorithm and feature-configuration selection were determined empirically through a systematic comparison across four tree-ensemble algorithms (LightGBM, Random Forest, XGBoost, CatBoost) and four feature configurations (64-dim, 320-dim, 584-dim, and a SelectKBest-reduced variant of the full vector), with results confirmed via a five-seed robustness check and a tuned session-aware GroupKFold comparison.

Table 3. Layer 2 model specifications.

Model	**Task**	**Training Protocol**	**Output to Layer 3**
Model 1 (Binary Detector)	Normal vs. anomalous	CatBoost on 80/20 rebalanced training partition (≈3,740 Normal + 935 Anomaly). F2-score tuning objective. StratifiedGroupKFold by session. Isolation Forest unsupervised baseline. Model-lock gate: if recall < 0.50 AND F1 < 0.35, pipeline stops; automatic fallback to next-best algorithm. In the final run, gate did not fire.	Binary label + confidence score
Model 2 (Fault Classifier)	11-class fault identification	CatBoost on all anomalous training samples (≈935 windows across 11 classes including Jamming). Class weights from actual post-split counts. RandomizedSearchCV with F1-macro. StratifiedGroupKFold by session. Two smallest classes ($n \approx 37$, $n \approx 43$) use stratified 5-fold CV with ±95% CI.	Fault type label + per-class F1
SHAP TreeExplainer	KPI attribution and explanation grounding	Applied post-hoc to both models. Yields top-K per-channel Shapley values with direction and magnitude. No GPU required.	Top-3 SHAP channels with direction and magnitude

The cascade is evaluated via a full 12-class confusion matrix computed over all held-out samples, correctly accounting for Model 1 false positives and negatives. A conditional diagnostic metric (Model 1 recall × Model 2 accuracy on true anomalies) additionally isolates per-stage error sources.

6.6 Layer 3: RAG Pipeline and Standards-Grounded Explanation

6.6.1 Alignment Table Construction

The Fault-to-Standard Alignment Table is the primary research artifact of this project. It maps 10 of the 11 TelecomTS fault types to specific 3GPP Technical Specification clauses and O-RAN components. Jamming is excluded: as a physical RF attack, it has no corresponding 3GPP specification clause. For Jamming events, Layer 3 falls back to ticket-retrieval-only generation without alignment table lookup.

Construction methodology. The Week 1 go/no-go check confirmed that all 10 synthetic TelecomTS fault types could be supported by the GSMA knowledge-graph evidence, with 6 directly supportable and 4 supported at the mechanism level. Each alignment row was constructed in three stages: (1) a candidate 3GPP clause was identified through keyword-based search of the GSMA `telecom-kg-rel19` knowledge graph; (2) a candidate O-RAN component was identified through keyword-based search of the GSMA `oran_spec_knowledge_graph`; and (3) the final citation was verified against the primary 3GPP/O-RAN specification text, with the document ID, release version, clause number, and evidence span recorded.

Hardening through audit. Early KG-derived candidates were refined during primary-specification verification and the technical audit (§8.4.1). Notably, the Doppler Shift row was corrected from TS 38.901 to **TR 38.901**, a Technical Report (informative study document) rather than a Technical Specification (normative); the citation validator now accepts both TS and TR prefixes. The final canonical table is shown in Table 4.

Table 4. Fault-to-Standard Alignment Table.

Fault Type	**3GPP Reference**	**Clause**	**O-RAN Component**
Antenna Failure	TS 38.141-1	§4.2.1	O-RU
Buffer Overflow (Gradual Buildup)	TS 38.321	§6.1.3.1	O-DU
Co-Channel Interference (Mild)	TS 38.141-1	§7.8	O-RU
Co-Channel Interference (Severe)	TS 38.141-1	§7.8	O-RU
Doppler Shift (Severe)	**TR 38.901**	§7.6.6	O-DU / O-RU
Faulty Handover Algorithm (Too Frequent)	TS 38.133	§4.3.5	Near-RT RIC
Faulty RF Filters (Temporal)	TS 38.141-1	§7.5	O-RU
High Network Congestion (Gradual Buildup)	TS 28.552	§5.1.1.2.1	O-DU
High Network Congestion (Sudden Spike)	TS 28.552	§5.1.1.2.1	O-DU
Resource Allocation Bugs	TS 38.321	§5	O-DU / Near-RT RIC
Jamming	*None (physical RF attack)*	N/A	N/A

Design notes. Faulty Handover Algorithm is attributed to the Near-RT RIC because handover decisions are orchestrated at the RAN Intelligent Controller layer in the O-RAN architecture, not at the O-RU or O-DU. The table serves two functions: prompt augmentation (the C3 condition of Track C) and citation validation (the expected reference per fault type in `validate_citation()`).

6.6.2 ChromaDB Ticket Retrieval

The troubleshooting tickets and QnA anomaly reasoning traces from anomalous training-partition samples are embedded using all-MiniLM-L6-v2, a 22M-parameter Sentence-BERT

model producing 384-dimensional dense embeddings, selected for its balance of embedding quality, inference speed (∼80 MB footprint), and zero licensing cost. These are indexed in a ChromaDB collection (cosine distance space, ∼51 MB on disk). Indexing is idempotent. **935 training tickets** are indexed.

Cosine similarity over L2 distance was a deliberate design choice: fault ticket descriptions vary substantially in length, and L2 distance scales with vector magnitude, penalizing verbose tickets describing the same fault pattern. Cosine similarity is length-invariant, measuring only angular distance. Retrieval converts ChromaDB distances correctly as similarity $= 1.0 -$ distance.

Query construction. For each anomaly event, a structured natural-language query is built from the Layer 2 output, encoding the fault type label, the top-3 SHAP-attributed KPI channels with their values and directions relative to the normal baseline, and key signal statistics.

Confidence gating. ChromaDB returns the top-5 most similar tickets. A threshold of **0.35** is applied: if all returned similarity scores fall below this value, the system raises a `low_confidence` flag. This flag propagates to the dashboard and to the LLM prompt, preventing the model from hallucinating context.

6.6.3 RCA Evidence Integration

A pre-computed RCA (Root Cause Analysis) evidence store (`rca_evidence.json`) contains 1,235 records, one per anomalous window, each holding three layers of evidence:

- **Layer A (Observational):** per-channel KPI statistics for the window (mean, standard deviation, minimum, maximum for all 18 channels).
- **Layer B (Model Attribution):** SHAP feature attributions identifying which KPI features most strongly drove the classifier's prediction, each with channel, SHAP value, effect direction, and comparison against the normal baseline mean.
- **Layer C (Domain Standards):** the causal mechanism description for the predicted fault type, the applicable 3GPP/O-RAN reference, and the O-RAN component.

An `RCALoader` class loads the store at initialization for O(1) lookup, and provides a prompt-ready context string constrained to under 400 tokens. For Jamming predictions, a note is included that no 3GPP clause applies.

RCA evidence is injected **exclusively into the C3 prompt condition** of the Track C ablation; for C1 and C2, the RCA context is set to an empty string, preserving the experimental design.

6.6.4 LLM Explanation Generation

Retrieved tickets, the RCA evidence context, and the alignment table row are fused into a structured prompt. The LLM is instructed to return exactly five fields in a JSON object: `root_cause`, `3gpp_reference`, `oran_component`, `recommended_action`, and `reasoning_trace`.

Citation validation applies two independent checks: (1) format regex, where the `3gpp_reference` field must match the pattern for 3GPP TS/TR numbering; (2) alignment table cross-check, where the extracted TS number must appear in the alignment table's known-valid reference list for the predicted fault type. A citation that passes format validation but fails the cross-check is flagged as "hallucinated-but-valid-format."

LLM configuration. Development used Ollama with Llama 3.1 8B on the Queen's University HPC cluster (Frontenac). The primary backend is **Gemini (`gemini-3.5-flash-lite`)** via the google-genai SDK, adopted for improved JSON output compliance, lower latency, and more generous rate limits; it completed all 90 Track C LLM calls. The Groq backend is retained as a configured fallback.

6.6.5 Evaluation Harness and Citation Validation

The evaluation harness separates format validity from evidence validity and checks the generated TS number against the fault-specific Alignment Table reference set. This makes it possible to distinguish a correctly grounded citation from a citation that merely has a valid-looking format.

The harness records parse failures, deterministic template fallbacks, retrieval confidence, and the evidence sources supplied to the LLM. The same validation logic is used across local development and the final demonstration environment.

6.6.6 Standards-Grounding and O-RAN Coverage Iterations

The initial project criterion was to establish support for at least 7 of the 10 synthetic fault types. Rather than treating a keyword match as sufficient evidence, the workflow was refined to distinguish direct support from mechanism-level support. This resulted in a final 10/10 supportability assessment.

For the O-RAN KG iteration, the extracted evidence set contained 15,103 nodes and 5,954 chunk nodes. The fault-specific search produced evidence counts for all ten synthetic categories, including 579 Antenna Failure matches, 531 Buffer Overflow matches, 419 each for mild and severe CCI, 121 Doppler Shift matches, 278 Handover matches, 1,160 Faulty RF Filters matches, 491 and 429 matches for the two congestion patterns, and 661 Resource Allocation Bugs matches.

6.7 Dashboard and Deployment

The final system output is a Streamlit dashboard deployed on Streamlit Community Cloud. The live deployment is accessible at: `https://kpi-rag-group15.streamlit.app/`

HuggingFace Spaces was the initial deployment target but was abandoned after a policy change requiring a paid PRO subscription; Streamlit Community Cloud provides free hosting with GitHub integration.

For each processed sample, the dashboard displays **six panels**: (1) Detection panel, showing fault type, confidence score, and binary verdict; (2) SHAP panel, with an interactive bar chart of top-3 KPI channel attributions; (3) KPI signal panel, with time-series plots of the top-3 channels across the 128-timestep window; (4) RCA Evidence panel, showing the three-layer evidence for the diagnosed window; (5) Explanation panel, with root cause, 3GPP clause, O-RAN component, and recommended action; (6) Sources panel, listing retrieved ticket IDs with similarity scores and the alignment table row used.

Heavy resources (embedding model, ChromaDB collection, RCA evidence store) are cached so they load once, which is essential within the free tier's 1 GB RAM limit. The deployed application operates within approximately **430 MB**.

7 Preliminary Results

This section presents the experimental journey through Phase 1, from initial experiments to the final validated pipeline. Results are presented in chronological order to show the progression of decisions and findings. All final metrics are from the final validated pipeline, using the corrected session-level holdout split and the fixed 584-dimensional feature array.

7.1 Feature Engineering and Scale Ablation

The initial hypothesis was that a 582-dimensional scale-preserving feature vector (global statistics + NME patchwise scale + first-order differences + categorical encoding) would outperform a statistics-only baseline. A series of experiments was conducted to test this hypothesis systematically.

Initial ablation results (window-level split, Random Forest baseline):

Table 5. Scale ablation results (window-level split, Random Forest baseline).

Config	Dims	Binary F1	Fault F1-macro	Samples/Feature
A: Statistics only	64	0.981	0.945	15.4
B: Stats + Patchwise Scale	320	0.975	0.894	3.1
C: Full vector	582	0.964	0.883	1.7

The full vector dimension was later corrected to 584 following the categorical feature fix. Results here are from the window-level split and presented solely to illustrate the ablation finding. Final results on the corrected session-level holdout split are in Section 7.4.

Key finding: Config A (64-dim statistics only) outperformed Config C on both tasks. The root cause was identified as the curse of dimensionality: with only 988 anomalous training samples and 582 features, the ratio is 1.7 samples per feature — well below the minimum of 10 samples per feature generally required for reliable tree-based classification. Config A at 64 dimensions gives 15.4 samples per feature, satisfying this constraint.

Investigation chain triggered by this finding:

Table 6. Investigation chain following the scale ablation finding.

Investigation	Hypothesis Tested	Finding	Decision
Investigation 1: Scale imbalance	Scale imbalance (TX_Bytes reaches 455M vs. RSRP at −70 to −141)	Selective log1p on 5 traffic channels had minimal impact. Config ordering unchanged.	Scale imbalance not the root cause
Investigation 2: Algorithm comparison	Algorithm-specific? (Maybe RF cannot exploit 582 features)	Config A won across all four algorithms (RF, XGBoost, LightGBM, CatBoost). Not algorithm-specific.	Curse of dimensionality confirmed
Investigation 3: SMOTE oversampling	Can oversampling fix the ratio?	SMOTE increased anomalous count ≈4×. B/C improved slightly but A still dominant.	Not sufficient to rescue Config C
Investigation 4: Feature selection (SelectKBest)	Are statistics the genuinely best 64 features?	KBest-selected 64 from 582 scored 0.831, worse than Config A (0.984).	Config A design confirmed
Investigation 5: Hyperparameter tuning	Can regularisation rescue Config C?	Tuned Config C (0.955 binary F1) still below untuned Config A (0.984). Tuning hurt both configs.	Fundamental constraint. Config A is optimal.

7.2 Algorithm Comparison and Final Model Selection

Four tree-ensemble algorithms were compared across four feature configurations on the corrected session-level holdout split. Results were checkpointed to CSV to avoid re-computation. A five-seed robustness check confirmed the winner was not a single-seed artefact.

Table 7. Algorithm comparison results (Config A, session-level holdout split).

Algorithm	**Binary F1 (Config A)**	**Note**
Random Forest	0.665	
XGBoost	0.533	
LightGBM	0.364	Winner in ablation series (window-level split) but not on corrected split
CatBoost	**0.802**	**Best algorithm.**

CatBoost was selected as the final algorithm. The five-seed robustness check on the originally favoured LightGBM found the 'Config A wins' margin was smaller than seed-to-seed variability. The comparison was re-run specifically on CatBoost; Config A was confirmed as the winner on a tuned, session-aware GroupKFold comparison.

Final configuration: CatBoost + Config A (64 dimensions).

Following the selection of CatBoost and Config A as the optimal algorithm and feature configuration, two consequential pipeline defects were identified and corrected before final evaluation: the categorical feature block was rebuilt from six zero-filled columns to eight properly encoded columns (expanding the feature vector from 582 to 584 dimensions), and the train/test split was redesigned from a row-level stratified split to a session-level holdout split to eliminate adjacent-window leakage. All results from Section 7.4 onwards are based on the corrected 584-dimensional feature vector and the session-level holdout split.

7.3 Pipeline Defects Found and Corrected

Two consequential defects were identified and corrected during pipeline development. Both were present in all prior pipeline versions and invalidated earlier reported results.

Table 8. Pipeline defects identified and corrected.

Defect	Impact	Correction
Dead categorical features (Columns 576–581)	Six columns reserved for protocol encoding were silently zero-filled across all 32,000 windows in every prior run. The protocol signal never reached the model.	Replaced with a real 8-column one-hot encoding (4 categories × 2 directions: TCP, UDP, None, EMPTY). Feature count corrected from 576 to 584. Validation gate added.
Data leakage in train/test split	Original row-level stratified split produced a near-duplicate rate of 31.8% (test windows with $\geq$0.999-cosine-similar training match) due to 75% adjacent-window overlap.	Replaced with session-level holdout split (300s gap threshold, 33 sessions). Near-duplicate rate reduced to 4.05%. Zero session overlap confirmed.

The impact of the split correction was clear: fixing the split alone, without changing the model or features, dropped Model 1 precision from 0.976 to 0.209, confirming that the early-phase metrics were an artefact of the leaky split, not real model performance.

7.4 Final Model Results

All results in this section are from the final pipeline, using the session-level holdout split and the corrected 584-dimensional feature array (of which Config A uses the first 64 dimensions).

Model 1: Binary Anomaly Detector

Table 9. Model 1 (binary detector) evaluation results.

Metric	Value	Context
Recall (natural test rate)	0.897	Primary operational metric. F2-score objective: missed faults are costlier than false alarms
Precision (natural test rate)	0.196	Direct consequence of recall-first tuning and 26:1 class imbalance
F1	0.698	Evaluated on N=1,000 subset with 80/20 normal/anomalous ratio
Isolation Forest baseline F1	0.227	Unsupervised baseline; Model 1 outperforms it by +47 percentage points
Confusion matrix (TN/FP/FN/TP)	4907 / 1105 / 31 / 269	Natural test rate. FN=31 missed faults; FP=1105 false alarms
Train F1 / Test F1 gap	0.976 / 0.321 = 0.655	Flagged. Driven by 26:1 class imbalance at natural rate, not overfitting

The session-level holdout split used in this evaluation is stricter than the standard row-level protocol, as it eliminates adjacent-window leakage. The reported F1 = 0.698 reflects this harder evaluation setting.

Model 2: Fault Classifier (11-class)

Table 10. Model 2 (fault classifier) evaluation results.

Metric	Value	Notes
Accuracy	0.677	11-class fault classification on TelecomTS test partition
F1-macro (single train/test split)	0.553	Optimistic point estimate. Falls outside CV range; reported for context only
F1-macro (5-fold session-aware CV)	0.333 ± 0.076	**Primary reported figure. 95% CI [0.267, 0.400]**
F1-macro with class grouping	0.354 ± 0.072	Evidence-based grouping: Co-Channel Mild/Severe and Congestion Gradual/Sudden merged
Train / Test F1-macro gap	0.994 / 0.553 = 0.441	Regularization mitigation tested; not effective (see Section 7.5)
Cascade correctness rate	0.647	Model 1 recall × Model 2 accuracy on correctly detected anomalies

The gap between the single-split estimate (0.553) and the cross-validated result (0.333) reflects the instability of single-split evaluation on a small anomalous test set. The cross-validated figure is the primary reported metric. The single-split estimate is included only for comparability with earlier pipeline versions evaluated without cross-validation. The limited anomalous sample count (1,235 windows across 11 fault types) is a known constraint of the current evaluation; future work will investigate synthetic data augmentation strategies to improve per-class generalisation.

7.5 Model 2 Mitigation Experiments

Table 11. Model 2 mitigation experiments.

Experiment	Single-split	CV result	Decision
Regularisation (reduced depth, increased L2)	0.553 → 0.508	Gap widened	Rejected. Removing capacity removed signal for rare classes without addressing root cause
SMOTE oversampling	0.553 → 0.593 (apparent gain)	0.333 → 0.319 (reversed)	Rejected. Apparent gain did not survive CV
Class grouping (Co-Channel Mild/Severe + Congestion pairs)	—	0.333 → 0.354 ✓	Adopted as optional reporting complement. Evidence-based merge from confusion matrix
Repeated 5-fold CV (3 seeds)	—	All three repeats produced identical fold partitions	Null result. Too few anomalous training sessions for shuffling to vary the partition

The mitigation experiments are summarized in Table 11. Regularization reduced the single-split score but did not address the underlying performance gap, while SMOTE produced an apparent improvement that was not sustained under cross-validation. The class grouping provided a small but consistent improvement in cross-validation performance and was therefore retained as an optional reporting complement. Repeated 5-fold cross-validation produced identical fold partitions across all three seeds because the number of anomalous training sessions was too small for shuffling to change the partition.

7.6 Key Finding: Implicit Catch-All Class

The most significant analytical finding of Phase 1 is a structural failure mechanism in the cascade, identified by examining Model 2 predictions on Model 1 false positives.

Mechanism: Model 2 is trained on eleven fault classes and has no way to express 'none of the above.' When Model 1 passes it a genuinely Normal window, Model 2 must choose one of its eleven labels and defaults to the class whose decision boundary sits closest to normal-looking telemetry. This destroys that class's precision on genuine test examples. The mechanism is distinct from small-sample scarcity: Co-Channel Interference (Mild)

Table 12. Catch-all class mechanism quantification.

Observation	Value
Model 1 false positives (Normal windows incorrectly flagged)	1,105
Fraction labelled "Co-Channel Interference (Mild)" by Model 2	82.2%
Co-Channel Interference (Mild) training support	≈126 samples (well above small-class threshold of 50)
Co-Channel Interference (Mild) precision on true test examples	0.00
Co-Channel Interference (Mild) F1 on true test examples	0.00

has adequate training support. It is documented in the exported model selection report for downstream RAG consumers.

7.7 SHAP Attribution Results

SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer on Model 2 across all anomalous test windows. The five globally most important features are shown in Table 13.

Table 13. Top-5 globally important SHAP features (Model 2).

Rank	Feature	Interpretation
1	`RSRP_max`	Peak signal strength; absolute scale is diagnostically informative
2	`RSRP_std`	Signal variability; characterises stability vs. disruption
3	`DL_MCS_mean`	Downlink modulation scheme; links signal quality to spectral efficiency
4	`UL_SNR_min`	Minimum uplink signal-to-noise ratio; worst-case signal quality
5	`UL_SNR_mean`	Mean uplink SNR; baseline signal quality level

All five top features are core link-quality indicators, consistent with telecom domain expectations. For 9 of 10 synthetic fault types, dominant SHAP channels are consistent with 3GPP-specified KPI effects. The one partial exception is High Network Congestion (Gradual Buildup), where RSRP dominates SHAP attribution despite PRB utilisation being the specification-expected primary metric, a correlation artefact in the synthetic injection rather than a true causal relationship.

These per-prediction SHAP attributions are exported as structured top-3 channel rankings with direction and magnitude, forming the retrieval query that Layer 3 uses to surface similar historical fault tickets from ChromaDB.

7.8 Implementation Iterations and Research Decisions

The preliminary implementation reflects iterative decisions at every layer of the pipeline. At the ML layer, the experimental history documented in Sections 7.1–7.7 shows how the algorithm, feature configuration, split methodology, and evaluation protocol each evolved from initial assumptions to empirically validated choices. The two defect corrections (Section 7.3) were enabled by the architectural decision to separate feature construction from modelling, with a machine-readable inter-layer schema defining the feature contract between the data preparation and modelling stages.

At the standards-grounding stage, a minimum go/no-go criterion of 7/10 supportable synthetic fault types was initially defined. Iterative KG extraction, keyword refinement, component matching, and primary-specification verification increased the validated coverage to 10/10. The resulting mappings were classified as either directly supportable or mechanism-level support, with confidence recorded per fault.

A second design iteration concerned the knowledge representation. A live Neo4j/Cypher knowledge-graph query path was considered but not selected for the current phase. The project adopted a static, validated Alignment Table as the standards-grounding interface, keeping standards evidence auditable and allowing the Track C experiment to isolate the contribution of explicit alignment-table grounding.

For the explanation layer, the evaluation design was tightened from simple reference-format checking to two-stage citation validation, separating format validity from evidence validity and checking the generated TS number against the fault-specific Alignment Table reference set.

8 Evaluation Plan and Executed Studies

Section 8.1 through Subsection 8.3.1 define the evaluation protocol, while Subsection 8.3.2 through Section 8.4 report the executed studies.

8.1 Track A: Signal Layer Detection Metrics

Table 14. Track A: signal layer detection metrics.

Metric	Definition	Baseline to Beat
Binary F1	Harmonic mean of precision and recall on the full test split, evaluated at natural 96.1%/3.9% distribution without resampling	TelecomTS binary detection task
Per-class accuracy (11-class)	Plain accuracy on the 11-class fault classification task	TelecomTS fault classification task
Per-class F1-macro	F1 reported per class and as macro-average; this is the first per-class F1 table for the 11-class TelecomTS task	No prior baseline; project contribution
Scale ablation ΔF1	F1 difference between 582-dim (full, with patchwise scale) and 320-dim (no patchwise), and 64-dim (statistics only)	Answers RQ1
Jamming generalization probe	279 Jamming samples evaluated as zero-exposure test cases after training on 10 synthetic types only	No prior baseline

Track A evaluates the signal-layer component of KPI-RAG from both overall detection and fault-level classification perspectives. The binary F1 measures the ability to distinguish normal from anomalous network states under the natural class distribution, while the 11-class accuracy and macro-F1 assess how reliably individual fault types are identified. The scale ablation examines whether the proposed 582-dimensional representation provides a measurable benefit over reduced representations, directly addressing RQ1. Finally, the Jamming generalization probe tests whether the detector can recognize a fault type that was completely excluded from training, providing an additional measure of generalization beyond the standard in-distribution evaluation.

8.2 Track B: Explanation Quality Metrics

Table 15. Track B: explanation quality metrics.

Metric	Definition	Target / Criterion
Citation faithfulness (O4)	% of generated explanations where the `3gpp_reference` field passes format regex AND alignment table cross-check	$\geq 70\%$ (O4 criterion)
Hallucinated-but-valid citations	% of explanations that pass format regex but fail alignment table cross-check	Report separately; target: $< 10\%$
Template fallback rate	% of explanations generated from deterministic template rather than LLM	Report separately; target: $< 15\%$
TeleQnA probe accuracy	Accuracy of the RAG pipeline on TeleQnA Lexicon and Standards Overview subsets	Report alongside citation faithfulness

Track B evaluates the quality and reliability of the explanations produced by the RCA layer. Citation faithfulness measures whether the generated 3GPP references are both correctly formatted and supported by the alignment table, while hallucinated-but-valid citations separately identify references that appear valid syntactically but are not supported by the established mappings. The template fallback rate measures how often the system relies on deterministic explanations instead of the LLM, providing an indication of the pipeline's ability to generate explanations dynamically. The TeleQnA probe provides an additional assessment of standards and telecommunications knowledge.

8.3 Track C: End-to-End Explanation-Layer Evaluation

8.3.1 Ablation Design

Track C evaluates the contribution of alignment-table grounding to the explanation layer through a controlled three-condition ablation. The conditions progressively introduce additional sources of evidence while keeping the evaluated windows and retrieval results consistent.

- **Condition 1: Label only.** LLM prompted with fault type label and signal statistics only. No ticket retrieval, no RCA evidence, no alignment table lookup.
- **Condition 2: Tickets + label.** LLM prompted with fault type label, signal statistics,

and top-5 retrieved tickets. No alignment table, no RCA evidence.

- **Condition 3: Full pipeline** (label + tickets + RCA evidence + alignment table): the complete Layer 3 pipeline as described. Primary condition under evaluation.

The primary outcome metric is citation faithfulness under each condition. If Condition 3 measurably outperforms Conditions 1 and 2 on citation faithfulness, the alignment table grounding contribution is empirically confirmed.

8.3.2 Executed Ablation Results

Setup. The ablation was executed over 30 windows (10 fault types × 3 samples per fault type) drawn from the test partition, with Jamming excluded. `random.seed(42)` was applied before sampling. The same 30 windows were used across all three conditions, and ChromaDB retrieval was performed once per window and shared across conditions. RCA evidence context was injected exclusively into Condition 3 (empty string for C1 and C2), preserving ablation isolation.

Table 16. Track C ablation results.

Condition	Valid Citations	Total	Citation Validity	Template Fallbacks
C1: Label only	2	30	**7%**	0
C2: Tickets + label	3	30	**10%**	1
C3: Full pipeline (RCA evidence + alignment grounding)	27	30	**90%**	1

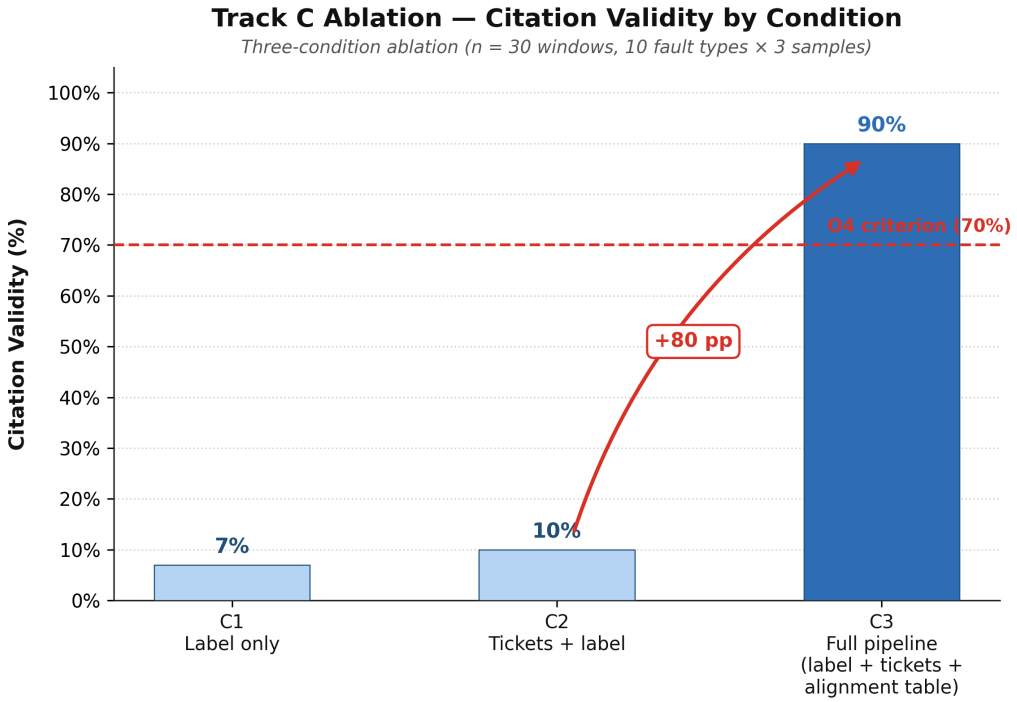


Figure 4. Track C ablation results: citation validity by condition ($n = 30$ windows). The +80 percentage-point improvement from C2 to C3 isolates the contribution of alignment-table grounding over ticket retrieval alone, confirming RQ2.

$\Delta(\text{C2} \rightarrow \text{C3}) = +80$ percentage points. $\Delta(\text{C1} \rightarrow \text{C3}) = +83$ percentage points. Condition 3 exceeds the O4 criterion ($\geq$70% citation faithfulness); Conditions 1 and 2 fall far below it.

The C2 fallback was a parse/retry failure; the C3 fallback occurred at sample 21 (CCI Severe), where three consecutive API calls returned HTTP 429 (rate limit exceeded). The resulting deterministic template fallback was correctly counted as a citation validity failure, yielding $27/30 = 90\%$.

Experimental controls.

- Identical windows across conditions, with retrieval shared per window.
- Prompt contamination prevention: an earlier iteration of the C1/C2 prompts included a format hint that inadvertently boosted baseline citation rates; it was removed.
- Real window diversity: an earlier run reused one synthetic payload per fault type; this was fixed to use unique real windows.
- Template fallbacks counted as failures in the citation validity metric.

Honest interpretation. The C3 citation validity metric measures prompt-compliance: given the correct 3GPP standard explicitly provided in structured context, does the LLM incorporate it correctly? It does not measure the LLM's independent knowledge

of 3GPP standards; C1 at 7% establishes that baseline. The accurate scientific claim supported by the evidence is: **when the correct 3GPP standard is provided in structured context, the LLM incorporates it correctly in 90% of cases.** The +80 pp improvement from C2 to C3 isolates the contribution of alignment-table grounding over retrieval alone, answering RQ2 affirmatively.

8.4 Explanation-Layer Audit, Reliability, and Test Suite

8.4.1 Technical Audit: Explanation Layer

A structured technical audit identified three substantive defects, all remediated:

Defect A (P0): Global citation validation. The initial `validate_citation()` accepted any correctly-formatted 3GPP reference for any fault type, inflating evaluation results. **Remediation:** refactored to fault-specific alignment lookup. Four negative tests added to prevent regression.

Defect B (P0): Incorrect document type for Doppler Shift. The alignment table initially mapped Doppler Shift (Severe) to `TS 38.901`; the correct reference is `TR 38.901`. All Doppler validation checks were failing silently. **Remediation:** corrected to TR 38.901 §7.6.6; format regex verified to accept both TS and TR prefixes.

Defect C (P1): Cosmetic RCA evidence integration. `rca_evidence.json` was displayed in the dashboard but not injected into the LLM prompt, a coherence failure. **Remediation:** `RCALoader` implemented; evidence wired into Condition 3 prompt construction.

8.4.2 Rate Limiting

The initial retry logic used fixed linear backoff (30/60/90 s), which proved insufficient during evaluation-scale runs (90 sequential Track C LLM calls). A `RateLimiter` was implemented with two complementary mechanisms:

- **Preventive:** a minimum inter-request interval (default 2.0 s, configurable) enforced between consecutive API calls.
- **Reactive:** on a 429 response, exponential backoff with jitter: wait = $\min(\text{base} \times 2^{\text{attempt}} + U(0,2), \text{max_wait})$, with base = 5 s and max_wait = 60 s.

8.4.3 Test Suite

The project maintains **118 tests across 16 modules (81 unit tests and 37 integration tests), all passing at 100%.**

Table 17. Key negative test cases (added after the audit).

Test	What it verifies
Wrong-fault reference rejection	TR 38.901 (Doppler ref) fails validation for CCI Mild
Hallucinated format rejection	TS 99.999 fails format check
Empty-string rejection	Empty string fails both validation checks
Generic wrong-TS rejection	TS 38.331 (RRC spec) fails for CCI Mild
Unrelated-query retrieval	"Chocolate cake recipe" returns similarity < 0.5
Jamming exclusion	Jamming not included in the Track C sample set

8.4.4 Configuration and Reproducibility

All system parameters are read from `configs/config.yaml`; no values are hardcoded in source files. `random.seed(42)` is applied in all contexts involving randomness, including Track C sample selection. Train/test split indices are pre-computed and stored as `.npy` files, ensuring consistent partitioning across runs and environments.

9 Expected Outcomes / Deliverables

- **Alignment Table:** Validated Fault-to-Standard Alignment Table: 10 rows linking TelecomTS fault types to 3GPP TS clauses and O-RAN components, with independently verified evidence spans. Primary research artifact.
- **Per-class F1 table:** Per-class F1 table for the 11-class TelecomTS fault classification task, the first such table in the published literature.
- **Scale ablation:** Scale ablation results: F1 comparison across 64-dim (statistics only), 320-dim (statistics + patchwise scale), and 584-dim (full corrected vector) feature configurations, plus a SelectKBest-reduced variant (Section 7.1).
- **Track C ablation results:** citation faithfulness under 3 conditions (label only / +tickets / +alignment table), directly testing the novelty claim; executed, (Section 8.3.2).
- **G-Eval pilot results:** Explanation quality will be evaluated using G-Eval, an LLM-as-a-judge framework proposed by Liu et al. (15). Four project-defined dimensions, Citation Validity, Fault Specificity, Actionability, and Causal Soundness, are evaluated by GPT-4 on 30 stratified samples, with 3 samples per synthetic fault type and Jamming excluded, using a 1–5 scale. Following the G-Eval scoring procedure, the final score is computed as a probability-weighted sum over the probabilities of the possible score values..
- **Dashboard:** End-to-end Streamlit dashboard deployed on Streamlit Community Cloud

with 6 panels (detection, SHAP, KPI signal, RCA evidence, explanation, sources).

- **Codebase:** Full codebase: modular Python implementation, shared JSON inter-layer schema, reproducible training scripts, evaluation harness, O-RAN alignment evidence artifacts, and dashboard implementation. Hosted on GitHub.

10 Timeline

Table 18. Project timeline (15 weeks).

Phase	Task	Weeks
1	Literature review, dataset evaluation, alignment-table feasibility scoping, proposal drafting and submission	1–3
2	Architecture and pipeline design with evaluation harness; alignment-table draft against 3GPP source documents; inter-layer schema. Checkpoint Wk5: Alignment Table draft complete	4–5
3	Core implementation: preprocessing, CatBoost detection/classification + SHAP, RAG pipeline, LLM generation, and full integration. Checkpoints: Wk6 table validated; Wk7 CatBoost trained; Wk8 Layer 1+2 integration; Wk9 RAG returning results; Wk11 full end-to-end demo	6–11
4	Evaluation: detection metrics, G-Eval pilot results, groundedness and ablation studies	12–13
5	Analysis and writing: results, limitations, final report, and consolidation of reproducibility artifacts	14–15

10.1 Iterative Milestones

- **ML pipeline:** Began with a 582-dimensional scale-preserving feature vector and window-level stratified split; progressed through a five-investigation ablation study confirming Config A (64-dim statistics) as the optimal configuration; identified and corrected two consequential pipeline defects (dead categorical features and adjacent-window leakage); and arrived at a validated CatBoost cascade evaluated on a session-level holdout split with cross-validated F1-macro reporting.
- **Standards coverage:** Started from an initial go/no-go criterion of at least 7/10 supportable fault types, refined through iterative evidence validation to reach 10/10 synthetic fault types supportable.
- **O-RAN evidence:** Began with KG extraction and fault-specific matching, progressed through direct and mechanism-level classification, and arrived at a confidence-aware evidence table.

- **Grounding validation:** Advanced from format-only citation checking to format plus fault-specific Alignment Table cross-check, with separate reporting of hallucinated-but-valid-format citations.
- **Architecture:** Live KG querying was considered and set aside in favour of a static validated Alignment Table, keeping evidence auditable and the evaluation controlled.

11 Risks and Mitigation

Table 19. Risk register and mitigation strategies.

Risk	Likelihood	Impact	Mitigation
Error cascade: Model 1 errors propagate to Model 2 and Layer 3, degrading combined metrics even if RAG is correct	High	High	Evaluate Model 1 and Model 2 independently before chaining. Run Layer 3 in two modes (ground-truth labels vs. predicted labels). Confidence gating: if confidence < 0.45, retrieve for top-2 fault hypotheses
LLM hallucination: plausible-format but incorrect 3GPP clause citations	High	Medium	Two-check citation validation (format regex + alignment table cross-check). Hallucinated-but-valid-format citations counted separately. TeleQnA probe accuracy as independent hallucination proxy
Congestion-Buffer confusion impacts Track C ablation	High	Medium	Report overall and per-class citation faithfulness in Track C
Cloud LLM API governance approval and rate limits	Medium	Medium	**Resolved**: migrated to Gemini (`gemini-3.5-flash-lite`) as primary backend; Groq retained as fallback; `RateLimiter` added (§8.4.2)

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Risk	Likelihood	Impact	Mitigation
Small rare-class test sets (Congestion-Sudden $n \approx 10$ test samples)	Certain	Low	Stratified 5-fold CV with ±95% CI. Results explicitly labelled as directional
Adjacent-window near-duplicate leakage across partitions	Certain (under row-level split)	High	**Resolved**: session-level holdout split adopted; near-duplicate rate reduced from 31.8% to 4.05%; zero session overlap confirmed
Alignment table KG coverage gaps for rare fault types	Low	Medium	All 10 rows confirmed against primary specification text. Template fallback available where evidence is insufficient

12 Resources Required

12.1 Compute

CatBoost training and SHAP attribution are CPU-bound and require no GPU; SHAP TreeExplainer runs entirely on CPU. LLM inference and iterative prompt and RAG development use Queen's University HPC cluster (Frontenac). Requested node specification: NVIDIA A100 40 GB (preferred) or V100 32 GB (minimum), 8 or more CPU cores, 64 GB RAM, and 100 GB storage for the full 15-week project duration.

12.2 Software

All software is open-source and freely available: Python 3.10+, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, sentence-transformers, ChromaDB, Ollama (development LLM), Groq API (fallback LLM), google-genai SDK (primary LLM backend), Streamlit, HuggingFace Transformers, pandas, numpy, matplotlib. No commercial licenses required.

12.3 Datasets

All four datasets are publicly available with no access request, payment, or data use agreement required for research use: TelecomTS (MIT license), GSMA telecom-kg-rel19 (CC BY-NC 4.0), GSMA oran_spec_knowledge_graph (CC BY-NC 4.0), TeleQnA (public on GitHub). Estimated total download size: approximately 5–10 GB.

12.4 Phase 2: Extended Evaluation

Phase 2 extends the evaluation through four complementary activities: the G-Eval pilot, the TeleQnA probe, re-evaluation using operator data and synthesized data, and comparison of SHAP-derived KPIs against the expected KPIs defined in the alignment table.

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